

Absolute Value Inequalities: Special Cases and Interval Notation

TEACHER KEY | Algebra II Honors | Unit 2 | Day 8 | TEK A2.6.F

Objective: I can write absolute value inequality solutions in interval notation, and recognize both special cases: no solution, and all real numbers.

Vocabulary

No solution — an isolated absolute value set $<$ or $\leq$ a negative number.

All real numbers — an isolated absolute value set $>$ or $\geq$ a negative number - true no matter what x is.

Part 1 — Interval Notation Reference

Inequality	Compound form	Interval notation
$ x > 5$	$x < -5$ or $x > 5$	$(-\infty, -5) \cup (5, \infty)$
$ x \geq 5$	$x \leq -5$ or $x \geq 5$	$(-\infty, -5] \cup [5, \infty)$
$ x < 5$	$-5 < x < 5$	$(-5, 5)$
$ x \leq 5$	$-5 \leq x \leq 5$	$[-5, 5]$

Part 2 — Variable on Both Sides

Example. $|3x + 1| - 4 \leq 2x - 1$

Isolate: $|3x + 1| \leq 2x + 3 \rightarrow -(2x+3) \leq 3x+1 \leq 2x+3 \rightarrow \mathbf{[-4/5, 2]}$

Part 3 — Two Special Cases, Side by Side

A. No solution. $4 + |x + 1| \leq 2$

Isolate: $|x + 1| \leq \mathbf{-2}$ An absolute value can never be **less than a negative number**. Solution: **no solution**

B. All real numbers. $4 + |x + 1| \geq 2$

Isolate: $|x + 1| \geq \mathbf{-2}$ An absolute value is ALWAYS **greater than or equal to a negative number**. Solution: **all real numbers, $(-\infty, \infty)$**

These two came from the SAME left side, just a flipped inequality symbol. In one sentence, explain why one has no solution and the other is always true.

Sample: **the range of $|x+1|$ is $[0, \infty)$, which misses every negative target for $\leq$, but satisfies every negative target for $\geq$**

Practice

- Solve $|x - 2| + 6 \leq 1$ **no solution**
- Solve $|x - 2| + 6 \geq 1$ **all real numbers**
- Solve $|2x - 1| > x + 5$ **$x < -4/3$ or $x > 6$**

Exit Ticket

Solve both $3 + |x - 4| < 1$ and $3 + |x - 4| > 1$. Explain in one sentence why they come out so differently despite looking almost identical.

first: $|x-4| < -2$, impossible, no solution. Second: $|x-4| > -2$, always true, all reals - the only difference is beating a negative target vs. never failing to beat one